Supporting Information

DMPNN Performance Estimates Diverge across Evaluation Boundaries in a Cyclic-Peptide Permeability Benchmark

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Scope and analysis boundary

This Supporting Information reports curation and split details, post-confirmatory robustness and sensitivity analyses for the frozen H1 comparison, a primary-source comparison with prior evaluation designs, the protocol timeline, deviations from the original split ladder, and the public-release inventory. Completed sensitivity analyses used the already accepted D0 out-of-fold predictions, checkpoint-only D3 residual reconstruction, or outcome-blind graph and curation inputs. No model was fitted, tuned, selected, replaced or recalibrated for those analyses. The internally prespecified 10,000-replicate block/seed bootstrap remains the inferential analysis for H1.

The H1 comparison used 6,895 curated records, 41 data sources, 18 frozen joint source/analogue blocks and five scheduled seeds. Joint-block and molecule-random prediction artifacts were verified against their frozen size and SHA-256 manifests before calculation.

The original version-2 split ladder listed molecule-grouped pseudorandom, chirality-aware Murcko, source-only, analogue-only and joint boundaries. Only the molecule-grouped and joint D0 arms were completed before outcome review. On 10 August 2026, source-only and analogue-only manifests were re-frozen operationally, and a newly added size-profile-matched molecule-random control was frozen, before any fit in those arms. Their pending results will be post-confirmatory and cannot alter the original H1 or H2 decisions. The Murcko arm remains uncompleted.

Curation accounting and fold assignment

The raw PAMPA table contained 7,298 rows. A row was classified as irrecoverably censored when either source detection-limit field was populated; 372 rows met this frozen rule. The remaining 6,926 rows had parseable structures and sequences. They were grouped by source and isomeric canonical SMILES. Multi-row groups were collapsed only when year, database version, topology signature and canonical sequence agreed across uncensored rows; the analytical endpoint was their median. Exactly 31 compatible groups contained more than one eligible row, explaining the arithmetic difference between 6,926 usable raw rows and 6,895 curated records. There were no incompatible groups and no invalid structure/sequence rows in the frozen population.

Table S1: Raw-to-curated accounting. The row-level curation manifest retains raw identifiers, censoring flags, inclusion or linked-exclusion status, and the final curated identifier.

Curation stage	Rows or groups	Rule
Raw PAMPA rows	7,298	Starting public table
Irrecoverably censored rows	372	Either detection-limit field populated
Uncensored usable rows	6,926	Parseable structure and sequence, numeric endpoint
Compatible multi-row groups collapsed	31	Same source and isomeric structure; compatible metadata; median endpoint
Curated source–structure records	6,895	Primary analytical population
Unique stereochemical molecules	6,862	`molecule_id` assignment units

For the molecule-grouped pseudorandom comparison, all records sharing `molecule_id` remained in one fold. Molecule groups were sorted by decreasing size with stable structure-derived identifiers breaking ties, then placed into the currently smallest of five folds. Each fold contained 1,379 records. Complete sources and analogue components were allowed to cross this comparison boundary. In the joint analysis, neither a source nor an analogue component could cross the outer boundary.

H1 block influence and aggregation sensitivity

Each frozen joint block was omitted in turn, and source-macro MAE was recalculated across the remaining sources for both evaluation arms. The difference was defined as joint-block MAE minus molecule-random MAE. This deletion analysis measures sensitivity to individual observed blocks; it does not define a new target population or confidence interval.

Alternative descriptive estimands assigned equal weight to records (row micro), sources (source macro) or blocks (block macro). For block-macro aggregation, source-specific MAEs were averaged within each block, blocks were weighted equally within each seed, and the five seed summaries were averaged.

Table S2: H1 aggregation sensitivity. Values are means across five scheduled seeds. The three weighting schemes estimate different target quantities and are reported descriptively.

Analysis	Joint-block MAE	Molecule-random MAE	Gap
Row micro	0.6430	0.2928	0.3502
Source macro	1.0046	0.4672	0.5373
Equal-block macro	0.9048	0.4386	0.4662

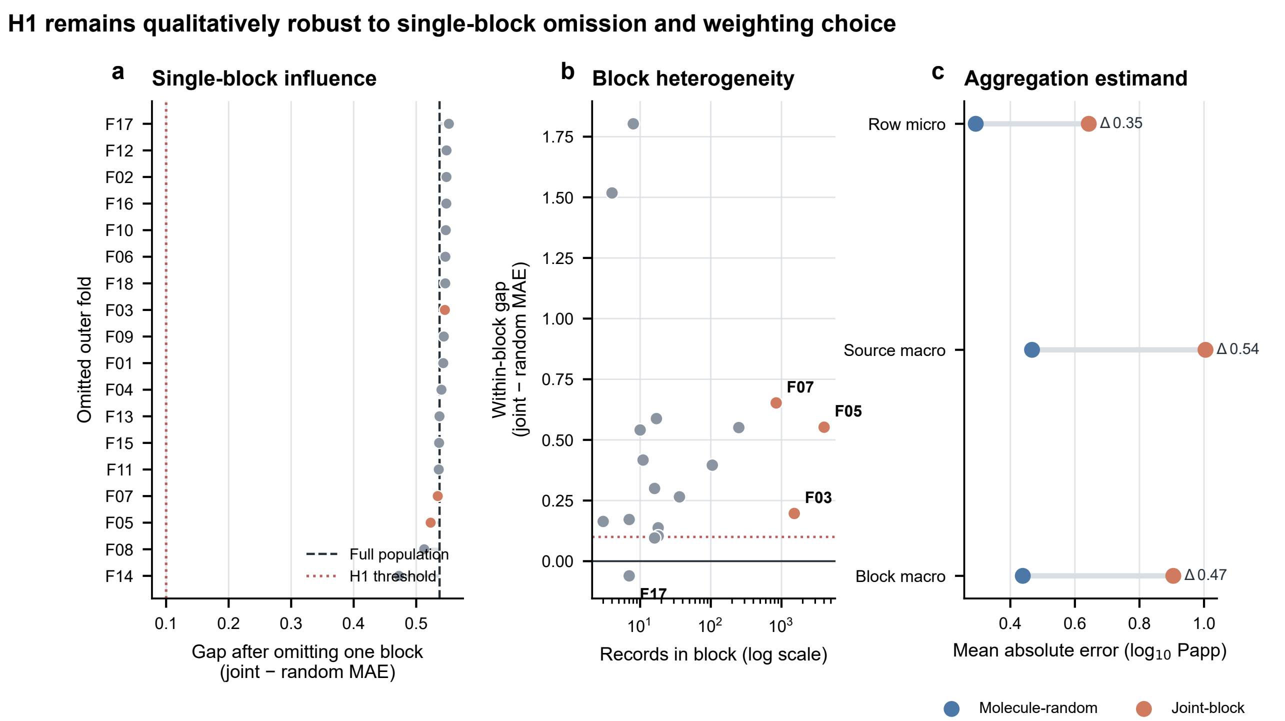

Figure S1: H1 influence and aggregation sensitivity. a, The D0 source-macro mean absolute error (MAE) gap between joint-block and molecule-random evaluation after omitting each of the 18 frozen joint blocks in turn. Points are ordered by the recalculated gap and labelled by the omitted outer fold. The black dashed line marks the full-population gap (0.537 log10 Papp units), and the red dotted line marks the preregistered 0.10 point-gap threshold. Orange points identify omission of one of the three largest blocks. **b,** Within-block source-macro gap versus frozen block record count for all 18 blocks. The three largest blocks and the sole block with a negative within-block gap are labelled. The x axis is logarithmic; the red dotted line marks a gap of 0.10. **c,** Five-seed mean D0 MAE under row-micro, source-macro and equal-block-macro aggregation. Blue and orange points denote molecule-random and joint-block evaluation, respectively, and connecting segments show the corresponding gap. All panels use the already frozen 6,895-record, 41-source prediction population across five scheduled seeds. These post-confirmatory analyses involved no refitting or model selection and are descriptive; the original block/seed bootstrap remains the inferential H1 analysis.

Small-cluster and high-leverage sensitivity

The five accepted seeds were first averaged within each source, giving 41 paired source-level joint-minus-random MAE differences nested in 18 sealed blocks. Three calculations then examined sensitivity to the small and unequal cluster set. An intercept-only CR1 variance estimator clustered the paired source differences by sealed block and used a t critical value with 17 degrees of freedom. A delete-one-block jackknife used the same t reference. Finally, all 2^18 assignments obtained by independently changing the sign of each block contribution were enumerated.

Table S3: Small-cluster sensitivity of the H1 source-macro gap. Delete-one-block point estimates ranged from 0.4724 to 0.5523. Seventeen of 18 block-specific mean gaps were positive, and 4 of 262,144 sign assignments were at least as extreme as observed. These analyses are post-confirmatory. CR1 and jackknife calculations treat the 18 blocks as independent clusters; the sign-flip test additionally assumes joint sign symmetry under the null. None replaces the internally prespecified block/seed bootstrap or creates additional independent regimes.

Sensitivity	Estimate	Standard error	95% interval or p value
CR1 block-clustered t interval	0.5373	0.0757	0.3777–0.6970
Delete-one-block jackknife t interval	0.5373	0.0756	0.3779–0.6968
Exhaustive block sign flip	0.5373	—	p = 0.00001526

Source-stratified error and point-prediction calibration

Absolute and signed errors were summarized within each source and seed, then averaged across the five seeds. Positive signed error denotes predicted permeability that is too high (less negative log10 Papp), whereas negative signed error denotes predicted permeability that is too low. Sources were retained regardless of size; extreme estimates from small sources are localization diagnostics rather than stable independent effects.

For point-prediction calibration, the five OOF predictions for each record were averaged first. Records were divided into ten equal-frequency prediction bins separately within each evaluation arm. Mean observed permeability was compared with mean predicted permeability in each bin. Calibration slope was obtained from a descriptive least-squares regression of observed values on mean predictions; no recalibration function was fitted or applied.

Table S4: Source-level bias and point-prediction calibration. Calibration used one five-seed mean OOF prediction per record. Prediction bins were arm-specific and contained all 6,895 records.

Calibration summary	Joint-block	Molecule-random
Calibration slope	0.5950	0.9637
Calibration intercept	-2.2192	-0.2075
Pearson correlation	0.3252	0.8729
Global signed bias	-0.1746	-0.0010
Decile-weighted absolute calibration error	0.1808	0.0269
Median absolute source bias	0.5779	0.0784
Sources with absolute bias above 0.5	23 of 41	3 of 41

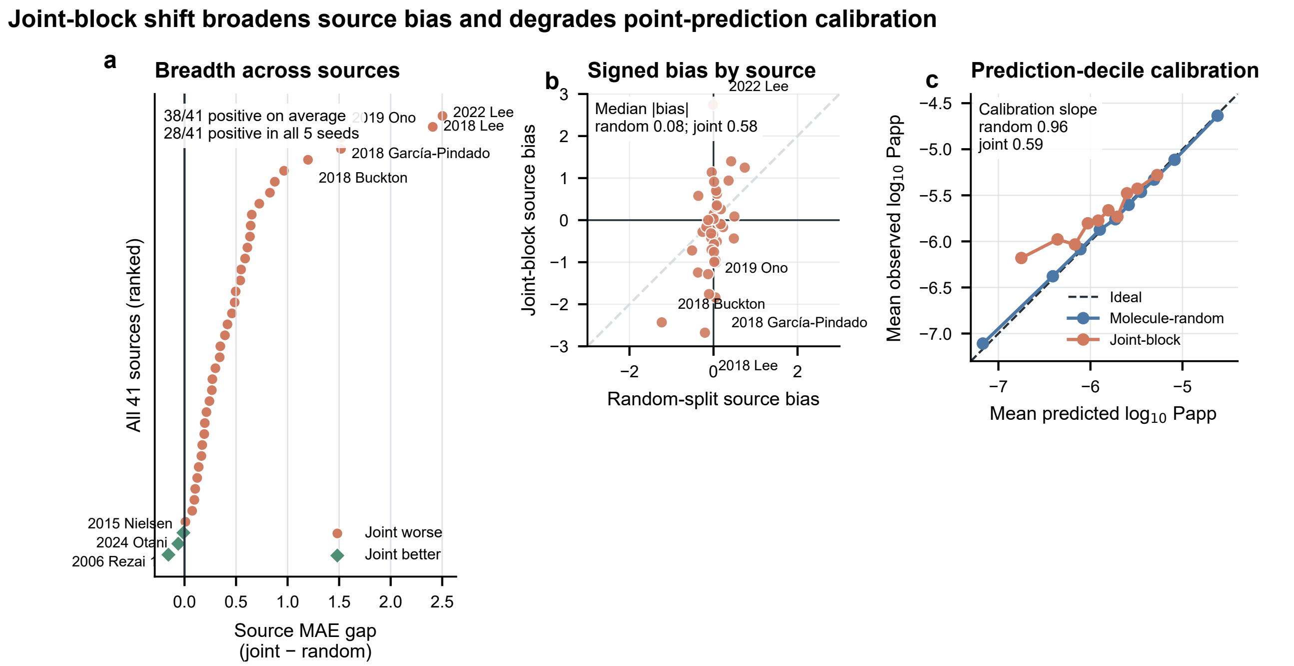

Figure S2: Source-stratified error and point-prediction calibration. a, Ranked source-level D0 MAE difference between joint-block and molecule-random evaluation for all 41 frozen sources. Orange circles indicate a higher joint-block MAE and green diamonds indicate a lower joint-block MAE; labelled points are the five largest positive differences and all three negative differences. Source summaries are means across five scheduled seeds. **b,** Five-seed mean signed error for each source under molecule-random and joint-block evaluation. Positive values indicate predicted permeability that is too high and negative values indicate predicted permeability that is too low. The dashed diagonal indicates equal bias under the two evaluation schemes; the five largest absolute joint-block biases are labelled. **c,** Point-prediction calibration after averaging the five frozen OOF predictions for each of 6,895 records. Points show mean observed versus mean predicted log10 Papp within ten equal-frequency, arm-specific prediction bins; the dashed diagonal is ideal calibration. Slopes are from descriptive least-squares regressions of observed on mean predicted values. No refitting, recalibration or model selection was performed. Source data are provided as CSV files.

Analogue-threshold sensitivity of evidence geometry

The outcome-blind analogue graph was reconstructed at the prespecified descriptive chiral ECFP4 Tanimoto thresholds of 0.70, 0.80 and 0.90. Fingerprint radius and length, chirality, the molecular-weight-ratio constraint, the exact one-token cyclic-edit relation, source provenance and all public curated records were held fixed. The threshold-0.80 reconstruction exactly reproduced every frozen primary component and block identifier.

Alternative thresholds were used only to compare graph and partition geometry. Existing OOF predictions were not rescored under these partitions because their training/test boundaries differ. A valid alternative-threshold performance comparison would require complete newly frozen nested retraining and would be exploratory after release of the primary outcomes.

Table S5: Outcome-blind analogue-threshold sensitivity. The effective block count is the inverse Simpson concentration index calculated from record shares.

Tanimoto threshold	Analogue edges	Analogue components	Joint blocks	Largest block share	Top-three share	Effective blocks
0.70	648,106	91	15	58.80%	93.02%	2.435
0.80	141,425	305	18	58.16%	92.39%	2.480
0.90	38,787	745	20	57.84%	92.07%	2.504

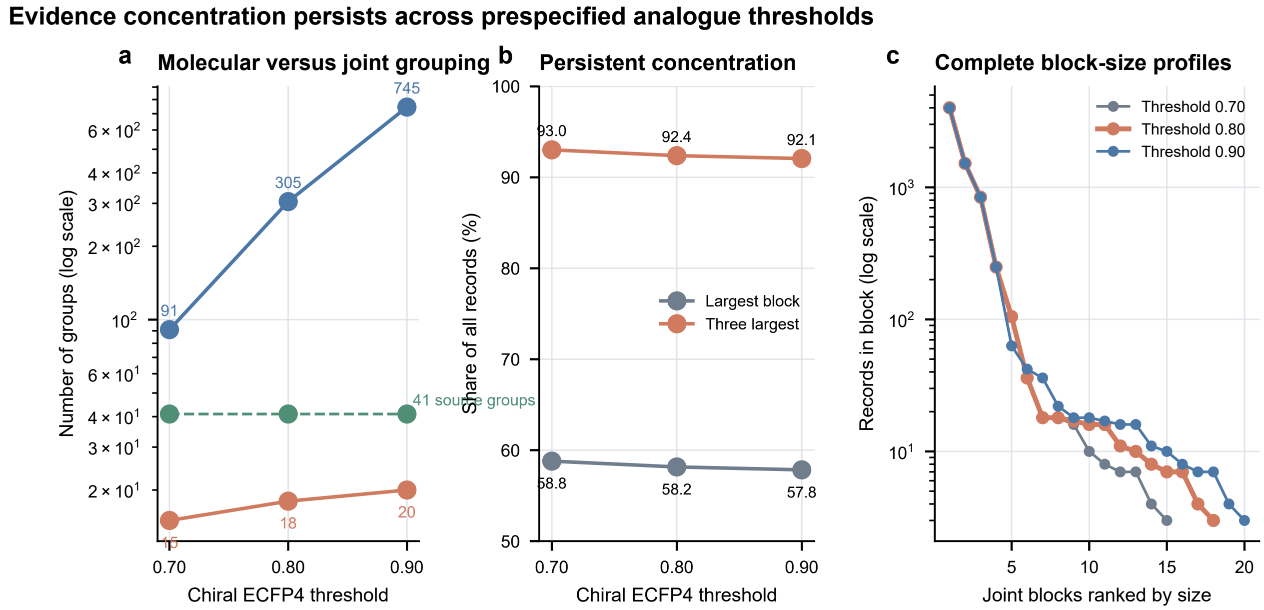

Figure S3: Analogue-threshold sensitivity of evidence geometry. a, Number of analogue components (blue) and source/component joint blocks (orange) reconstructed at chiral ECFP4 Tanimoto thresholds of 0.70, 0.80 and 0.90; the 41 source groups are shown as a green dashed reference. Counts use a logarithmic axis. The molecular-weight-ratio constraint, exact one-token cyclic-edit rule and source closure were held fixed. **b,** Percentage of all 6,895 records contained in the largest and three largest joint blocks at each threshold. **c,** Complete ranked joint-block size distributions on a logarithmic axis. The frozen primary threshold of 0.80 is emphasized in orange. Reconstruction used only public structure and provenance fields and exactly reproduced every primary 0.80 component/block identifier. No outcome, prediction or model-performance value was used. Alternative thresholds describe partition geometry only and were not used to rescore the frozen OOF predictions. Source data are provided as CSV files.

Prior-work comparison of evaluation boundaries

The comparison below was updated from primary reports through 10 August 2026. It is intended to delimit the present contribution, not to claim novelty for source holdout, chemical clustering, scaffold splitting, nested validation, or similarity-aware splitting individually.

Table S6: Primary-source comparison of evaluation boundaries. Geylan et al. is the closest provenance-aware precedent located. The present distinction is simultaneous closure of alternating source–analogue paths throughout model development. This is a search-limited comparison, not proof that no unindexed or unpublished study used a similar design. A post-confirmatory source-only, analogue-only and size-profile-matched random ladder has been frozen but is not yet reported.

Study	Reported evaluation boundary	Complete source held out?	Transitive analogue component held out?	Same boundary throughout nested development?
Geylan et al. (2024)	Four large external sources; separate source-group and canonical-group scenarios	Yes, in source arms	No	Partly; grouping scenarios were separate
CycPeptMP (2024)	Kennard–Stone holdout plus cross-validation	No	No	No
PeptideCLM (2025)	Six embedding/PCA clusters, leave one cluster out	No	No	Development within retained clusters, not joint blocks
MultiCycPermea (2025)	Random, Murcko/Jaccard-cluster OOD and permeability-cliff settings	No	No	No joint source–analogue nesting
13-method benchmark (2025)	Random and Murcko-scaffold splits; external set	No	No	No joint source–analogue nesting
DataSAIL (2025)	Generic similarity-aware one- and two-dimensional constrained splits	User-defined	User-defined	Depends on the user workflow
Present study	Molecule-grouped pseudorandom and joint source–analogue connected-component LOBO	Yes	Yes	Yes for the accepted joint analysis

Primary report identifiers: Geylan et al., DOI 10.1039/D4DD00056K; CycPeptMP, DOI 10.1093/bib/bbae417; Peptide-CLM, DOI 10.1021/acs.jcim.4c01441; MultiCycPermea, DOI 10.1186/s12915-025-02166-2; 13-method benchmark, DOI 10.1186/s13321-025-01083-4; DataSAIL, DOI 10.1038/s41467-025-58606-8.

Protocol timeline and public identifiers

Table S7: Protocol and release timeline. The complete chronology, file hashes and distinction between an internal freeze and an independently time-stamped public preregistration are provided in `PUBLIC_PROTOCOL_TIMELINE.md`. The public repository is https://github.com/schwarzeteeguo-creator/scaffoldseal-cp-evaluation; the remaining unresolved archival field is [Zenodo DOI pending]. Local hashes alone are not described as independent public preregistration.

Date	Protocol event	Public-release evidence
2026-07-17	Original v1 feasibility gate failed; compromised v1 final partition and vault retired	Failed protocol, decisions D-018/D-019 and repair audit retained
2026-07-17	Version-2 recovery map, CV governance and experiment plan frozen internally before any v2 fit	File hashes listed in `PUBLIC_PROTOCOL_TIMELINE.md`; no independent public timestamp
2026-07-17 to 2026-07-19	Pre-fit split, preprocessing, stopping and prediction controls reviewed	Test and audit records; training remained blocked until acceptance
2026-07-26 to 2026-07-27	H1 random-comparison plan accepted and executed	Accepted stage manifests, sealed OOF predictions and bootstrap outputs
2026-08-09	D123 sealed audit and one controlled metric release	Decisions D-052 to D-054 and H2 verdict report
2026-08-10	Zero-training sensitivity analyses and split-boundary completion plan frozen	Reporting scripts, input hashes and post-confirmatory internal freeze

Itemized Supporting Information and release mapping

Table S8: Mapping of SI statements to actual files. The authoritative package-level checklist is `RELEASE_INVENTORY_V0.7.md`. A manuscript statement must be removed or marked pending if its corresponding released file is absent.

SI item	Local/release files	Status in this version
Curation accounting	`curation_manifest_release_safe.csv`; `curated_group_manifest_release_safe.csv`; audit JSON; curation code and hashes	Included as release-safe manifests; structures and endpoint labels excluded
Primary joint split	outer record assignments, outer-fold manifest and inner-basket manifest	Complete and frozen
Comparison splits	molecule-grouped assignments; source-only, analogue-only and matched-size manifests	Outcome-blind manifests complete; corresponding model results pending
H1 headline	accepted joint/random metric files, OOF manifests and bootstrap output	Complete and frozen
H1 influence and aggregation	`analyze_h1_block_influence.py`; three CSVs; JSON summary; Fig. S1	Complete
H1 source calibration	`analyze_h1_stratified_calibration.py`; four CSVs; JSON summary; Fig. S2	Complete
Analogue-threshold geometry	`analyze_block_definition_sensitivity.py`; four CSVs; JSON summary; Fig. S3	Complete
H1 small-cluster sensitivity	`analyze_h1_small_cluster_sensitivity.py`; paired-source, cluster, deletion CSVs; JSON summary	Complete
Censoring and source metadata	`analyze_curation_source_audit.py`; release-safe manifests; source and note summaries; JSON	Complete; no raw structures or labels redistributed
D3 coverage aggregation	`analyze_d3_coverage_stratification.py`; per-seed, per-fold and per-source CSVs; JSON	Complete; checkpoint-only predictions, no fitting
Prior-work boundary comparison	`PRIOR_WORK_COMPARISON_NOTES.md`; `prior_work_evaluation_comparison.csv`	Complete through search date
Protocol and frozen files	v2 internal protocol, research map, CV governance, experiment plan, deviation table and public timeline	Complete locally; external timestamp absent
Public archive metadata	release checksum, `CITATION.cff`, license, GitHub URL and Zenodo DOI	Pending author/license/deposition actions

Censoring and source-metadata audit

The frozen curation rule excluded every row for which either detection-limit field was populated. A zero-training audit classified the two free-text fields conservatively and checked whether an excluded source-structure group was otherwise represented by an uncensored replicate. The classes below describe database annotations; they are not assumed to be mutually commensurate assay bounds.

Table S9: Censoring semantics and retained-group linkage. Note classes sum to 372. Numeric database entries among the excluded rows were heterogeneous: 270 were recorded as -10, 45 as -7, 15 as -4, and the remainder used other values. The raw table provided no harmonized structured columns for PAMPA pH, membrane composition, incubation time, temperature, or donor/acceptor conditions. Accordingly, the analysis neither substituted sentinel values as ordinary endpoints nor imposed a universal quantitative censoring bound. Source identity is treated as a combined provenance and protocol proxy.

Audit item	Rows or sources
Excluded rows carrying a detection-limit annotation	372
Raw-data sources containing at least one excluded row	23
Database-assigned or reported floor	259
Explicit upper limit	40
No reportable value	19
Not detected or below limit of detection	9
Solubility failure or not tested	7
Other detection-limit note	38
Excluded rows linked to a retained uncensored group	2
Excluded rows without a retained continuous-endpoint group	370
Completely censored source	1 source, 10 rows

Protocol completion and deviations

Table S10: Completion and evidential status of planned and added analyses. Source-only and analogue-only boundaries were listed in the original plan but were not completed before the joint and molecule-grouped outcomes were opened. Their operational re-freeze does not restore confirmatory status. The matched-size control was added only after recognizing that the completed H1 comparison changed both dependence and training-size profiles.

Planned element	Status before outcome review	Status in this version	Evidential role
Joint source/analogue D0	Completed	Reported	Primary internally prespecified evaluation
Molecule-grouped pseudorandom D0	Completed	Reported	Internally prespecified H1 comparison
Chirality-aware Murcko D0	Not completed	Pending	Originally planned secondary boundary; no result implied
Source-only D0	Not completed	Outcome-blind manifest re-frozen 10 August	Post-confirmatory when run
Analogue-only D0	Not completed	Outcome-blind manifest re-frozen 10 August	Post-confirmatory when run
Size-profile-matched molecule-random D0	Not in original ladder	Newly frozen 10 August; pending	Post-confirmatory training-size control
D1-D3 joint-block ladder	Completed	Reported	Internally prespecified H2 tests and ablations
Prospective matched-assay H3	Not performed	Pending external data	Future prospective confirmation

D3 coverage aggregation sensitivity

Each curated record contributed five algorithmic seed realizations to the primary slot-pooled coverage denominator. These realizations are useful for measuring execution variability but are not independent experimental observations. We therefore report the seed range and equal-source and equal-block descriptive estimands without a binomial confidence interval.

Table S11: Sensitivity of D3 empirical-interval coverage to aggregation. Slot pooling weights every record-seed pair equally; source and block macro estimates weight the 41 sources and 18 outer blocks equally, respectively. The wide block range reflects highly unequal and sometimes very small omitted regimes. All aggregations show under-coverage at the higher nominal levels.

Nominal coverage	Slot-pooled	Per-seed range	Equal-source macro	Equal-block macro	Block range
50%	44.1%	41.9-46.9%	31.9%	28.4%	0.0-80.0%
80%	67.2%	65.4-68.7%	58.6%	59.5%	0.0-100.0%
90%	77.0%	75.8-78.1%	71.4%	75.0%	0.0-100.0%

Source-data files

The influence package contains h1_leave_one_block_out.csv, h1_aggregation_sensitivity.csv and h1_per_block_effect.csv.

The stratified-calibration package contains h1_source_stratified_metrics.csv, h1_block_stratified_metrics.csv, h1_prediction_decile_calibration.csv and h1_calibration_summary.csv.

The threshold-sensitivity package contains block_definition_threshold_summary.csv, block_definition_ranked_sizes.csv, block_definition_partition_comparison.csv and block_definition_record_assignments.csv.

The small-cluster package contains h1_paired_source_effects.csv, h1_cluster_contributions.csv, h1_small_cluster_delete_one.csv and h1_small_cluster_sensitivity_summary.json.

The literature package contains prior_work_evaluation_comparison.csv and PRIOR_WORK_COMPARISON_NOTES.md.

The curation/source-audit package contains source_censoring_and_endpoint_summary.csv, censoring_note_class_summary.csv, censored_database_numeric_values.csv, curation_manifest_release_safe.csv, curated_group_manifest_release_safe.csv, curation_source_audit_summary.json and SHA256SUMS.

The D3 coverage-stratification package contains d3_coverage_by_seed.csv, d3_coverage_by_outer_fold.csv, d3_coverage_by_source.csv, d3_coverage_aggregation_sensitivity.csv, d3_coverage_stratification_summary.json and SHA256SUMS.